Sam Grant

(470) 955-2095 | sgrant41@gatech.edu | linkedin.com/in/samrgrant| Portfolio: sgrant41.github.io

Education

Georgia Institute of Technology *Expected Dec. 2025*

B.S. Mechanical Engineering

Relevant Courses: Machine Design, Composites Manufacturing, Bio-Inspired Design, Materials Selection/Design, Sustainable Design/Manufacturing, Surface Modeling & FEA, Fundamentals of Mechatronics, 2D/3D Dynamics

Experience

3D Print Lead/Prototyping Instructor, Invention Studio at Georgia Tech *Sep. 2021 – Present*

- Oversee fleet of 52 FDM & SLA printers and ~$110,000 budget, perform all intensive maintenance.
- Consult makerspace users on material/process selection and DFAM for research, academic, and personal projects.
- Spearheaded printer fleet replacement following cost-benefit analysis, secured funding and retrained 153 staff members on new procedures.
- Manage supplier relations for capital and consumables purchases.

Undergraduate Researcher, Dr. Amit Jariwala, Georgia Institute of Technology *Jan. 2023 – Jan. 2024*

- Planned and executed hands-on DFMA workshop for advanced manufacturing researchers in partnership with ETI Consortium and U.S. Dept. of Energy.
- Presented system to incorporate Design for Rapid Prototyping principles into makerspace staff training at International Symposium for Academic Makerspaces Oct 18-20, 2023.
- Designed process for recycling 3D prints at scale, characterized mechanical properties of recycled filament.
- Wrote curriculum for interdisciplinary design/prototyping course that was integrated into Georgia Tech's TechMade initiative for product realization.

Director of Operations, Invention Studio at Georgia Tech *Apr. 2023 – May 2024*

- Oversaw operations, hiring, and staff training at nation's largest volunteer student-run makerspace.
- Managed >150 staff members, scheduling shifts and setting hours of operations.
- Conducted avg. of 4 behavioral/technical interviews weekly for staff hiring and promotion.
- Implemented in-house shift management software, saving $5000 in annual expenses.
- Initiated executive board's development of tangible goals and 5-year strategic plan.

Lab Supervisor, Woodruff School of Mechanical Engineering *Aug. 2023 – Present*

- Teaching assistant for 247 students taking introductory product design/mechatronics course.
- Assist students in identifying failure points in mechanical systems, debugged Arduinos, sensors, motors, and pneumatics.
- Service equipment, including mills, lathes, laser cutters, and 3D printers; manage consumables inventory.

Technical Skills

CAE: SolidWorks CAD/FEA, NX CAD/FEA, Fusion CAD/CAM, Onshape, Ansys Granta, Rhino, Blender

Software: C++ (STM32 & Arduino), MATLAB, Magics, Simulink, Microsoft Office/Power Automate, 3DPrinterOS, Trello

Fabrication: CNC/Manual Mill, Manual Lathe, Waterjet, Laser Cutter, PCB Milling, Media Blaster, Sewing Machine, Woodworking, Metalworking, TIG Welding

Additive Manufacturing: DMLS (EOS M280), SLS (Formlabs Fuse 1+), SLA/mSLA (Formlabs Form 3/4, Nexa3D XiP), FFF/FDM (Ultimaker, Prusa, Bambu Lab), Markforged Continuous Fiber, Virtual Foundry Sintering

Projects

Competition Robot, Creative Decisions and Design *May 2023 – July 2023*

- Placed first in design review by panel of mechanical engineering faculty and staff.
- Led design team for autonomous competition robot.
- Created houses of quality, morphological charts, and evaluation matrices to quickly iterate through multiple solutions.

Drum Sander, Machine Design *Aug. 2024 – Dec. 2024*

- Final BOM cost in bottom 25^{th} percentile of peers.
- Designed support structure, power transmission, and housing for 1.2hp drum sander rated for 22,000 hours of use.
- Validated structural deflection requirements using FEA (NX Nastran).
- Produced ASME Y.14 drawings with appropriate tolerances for all components not off-the-shelf.
- Designed both V-belt and gear drive systems, compared transmission efficiency and total cost.